

INTRODUCTION TO ENGINEERING MATERIALS

→ According to Webster's dictionary, materials are substances of which something is made.

- ★ Engineering Materials:- parts of inanimate matter which is useful in the engineering profession.
- ★ Material Science:- It is the knowledge about properties, processing and complex relationship of materials.
- ★ Material Engineering:- It is the topic where we learn the knowledge to convert materials into usable demanding products. This basically bridges material knowledge from basic science to engineering disciplines.

Q.

↳

Why Material Science & Engineering is important to Engineers?

Ans:- Because through this topic engineers learn that-

- ~~The~~ A Mechanical Engineer needs high temp. materials to operate gas turbines and jet engines efficiently.
- An Electrical Engineer needs materials with faster operation and with minimum power loss.
- An Aerospace and Automobile Engineer needs material having very high (strength/weight) ratio.
- An Electronics Engineer requires materials useful in fabrication and miniaturization of electronic devices like chips, boards etc.
- Chemical Engineers requires material with low reactivity towards water i.e. Corrosion resistant materials.

★ Difference between "Material Scientists" And "Material Engineers".

→ A Material Scientist synthesises new materials whereas A Material Engineer is called upon to create products and systems using existing materials or to

develop techniques for material processing.

ENGINEERS OR APPLIED SCIENTISTS

Q. Why do we Study Material Science and Engineering?

Ans. Because -

- ① Material Selection is one key problem they face while working with materials in different branches like Meche., Electrical and Chemical.
- ② There are Several Criteria through which the right material is chosen -
 - First of all, the in-service conditions must be characterised. On only rare occasions a material does maximum and ideal combination of properties. Like a material having high strength lacks in ductility. So here it becomes a necessity to trade-off one condition for other condition. Thus compromise a reasonable compromise is necessary.
 - Second is Consideration of deterioration of material properties occur during its service. Example significant reduction in mechanical strength results from over exposure to corrosive environment.
 - Final point is the most crucial criteria and that is Consideration of economics. A material may have the ideal properties but if it is not economically bearable then that material is of no use in material science.

* Classification of Materials.

① According to General properties -

Ferrous

→ Metals with iron as a major component ($> 98\%$ Fe).

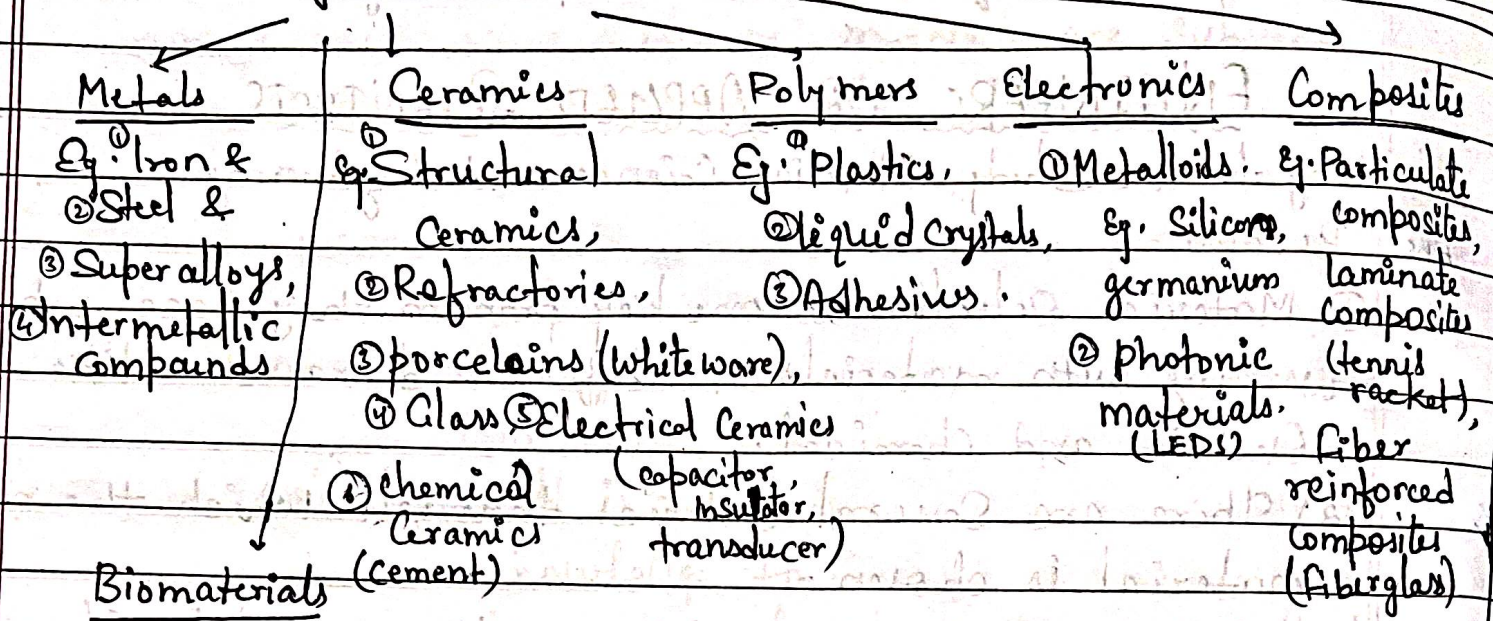
Eg. Cast iron, Wrought iron, Steel, Spring steel

Non-Ferrous.

→ rest all other metals and their alloys. Eg. Copper, Al, Zn, Pb, alloys like brass, bronze etc.

→ Non metals. Eg. Leather, rubber etc.

② According to Nature

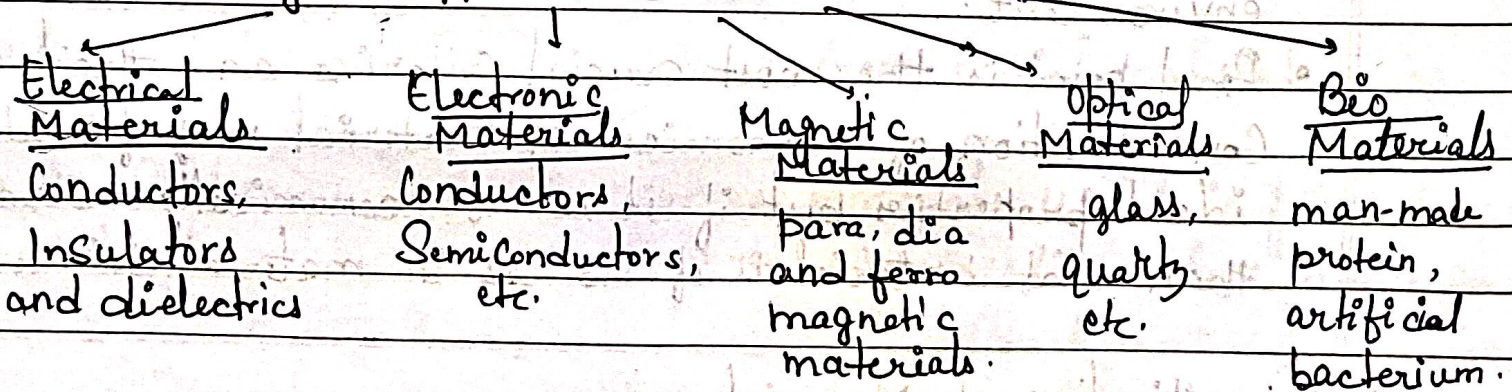


- ① Man-made proteins, (artificial bacterium)
- ② Biosensors etc.

Advanced & Smart materials

Eg. materials in Computer, optic fibres, (VCR, CD player)
 Space craft, rockets, Shape-memory alloy.
 piezoelectric materials, magnetostictive material
 MEM devices; Nanomaterials.
 Electro/Magneto rheological fluids.

③ According to Application.



Engineering Materials

Metals	Ceramics	Polymers	Composites	Electronic	Biomaterials	Adv. Smart
Ferrous Cast iron Carbon steel Alloy steel Stainless steel	Alumina, Diamond, magnesia, Silicone, Carbide, Zirconia.	Thermoplastic ABS, Acrylic, Nylon, polyethylene, polystyrene, vinyl.	Carbon fibre, Ceramic Matrix, Glass fibre, Metal Matrix.	Si, Ge, Photonic Material, lasers, LEDs.	Man-made proteins, Biosensors	piezoelectric Ceramics, Shape-memory Alloys, optic fibres, Nano metals, Electro rheological
Non-ferrous Al, Cu, Pb, Mg, Ni, Sn, Zn, Ti, brass, bronze etc.		Thermosetting Epoxy, phenolic, polyester.				
		Elastomer Butyl, Fluorocarbon Neoprene rubber, Silicone.				

Qualitative Analysis

Quantitative Analysis

- ① It identifies the presence or absence of a particular substance in sample.
- ② It is ~~performed~~ performed using simple chemical test that produce a visual change such as a colour or precipitation etc.
- ③ It is used to identify unknown substance to confirm the identity of a known substance
- ① It measures the amounts or concⁿ of a substance in a sample.
- ② It is performed using instrumental methods that produces numerical results such as mass or volume.
- ③ It is used to determine impurity or composition of a substance

Spectroscopy: Interaction of electromagnetic radiation with matter. it is the branch of science which deals with interaction of electromagnetic radⁿ with matter

Spectroscopy

Atomic

Molecular

• Spectrum:- The absorption of electromagnetic radⁿ by atoms & molecules causes electrons to get excited from lower to higher energy levels, when they de-excite, release quantised energy packets. this is spectrum.

• Spectroscopy
 → Atomic
 → Molecular

→ In atomic spectroscopy, only electronic transitions are involved, so Electronic spectra can be used to obtain detailed informatⁿ about ~~Structure of atoms~~ Structure of atoms.

→ But on Contrary, in molecules's case they absorb energy and not only shows electronic transitions but makes transitions b/w its rotational & vibrational states. that's why molecular spectra are more complicated

→ Spectroscopy is used for the absorption, emission and scattering of EM-radiation.

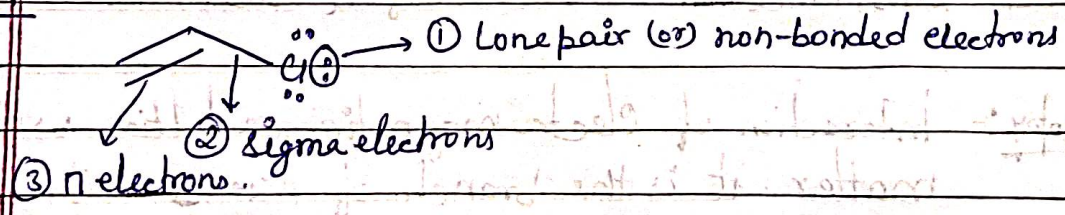
Spectroscopy
 ↙ absorption
 ↘ Emission
 ↘ scattering

Emission or Absorption
 ↙ Continuous
 ↘ Discontinuous
 ↙ line
 ↘ band

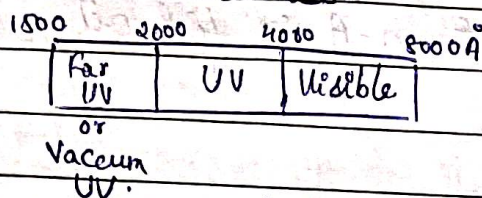
• UV- VIS spectroscopy (or) electronic spectroscopy (or) electronic excitation spectroscopy.

Q. what are different kinds of electrons in Vinyl chloride?

Soln:-



→ Electronic Spectroscopy involves the promotion of e^- from ground to higher energy states. It is very useful to measure the no. of conjugate double bonds ($\text{—}\text{=}\text{—}\text{=}\text{—}$) and also aromatic conjugation within the various molecules.



→ The region below 2000 Å is called vacuum UV region. The far UV region is not much studied due to absorption by oxygen and nitrogen. Moreover, studies in these region required vacuum instrument.

→ Types of excitations

$\sigma \rightarrow \sigma^*$
 $\pi \rightarrow \pi^*$
 $n \rightarrow \sigma^*$
 $n \rightarrow \pi^*$

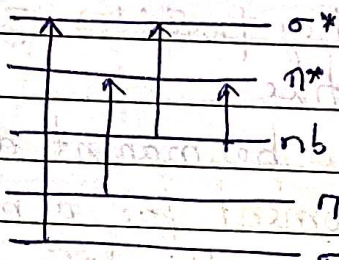
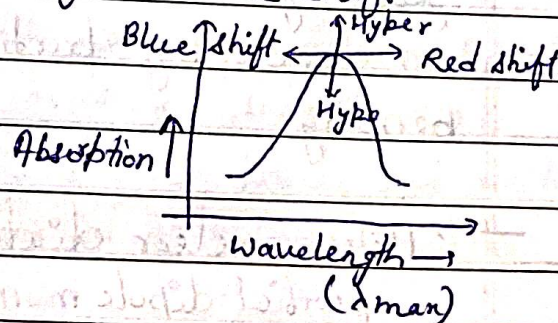


Fig. Energy level diagram of diff electronic transitⁿ.

Q. Short note on $\sigma \rightarrow \sigma^*$ transition

• Important terminology in UV Spectroscopy photometer

- ① Chromophore
- ② Auxochrome
- ③ Bathochromic shift (Red shift)
- ④ Hypsochromic shift (Blue shift)
- ⑤ Hypochromic shift
- ⑥ Hyperchromic shift.



• Benzene :-

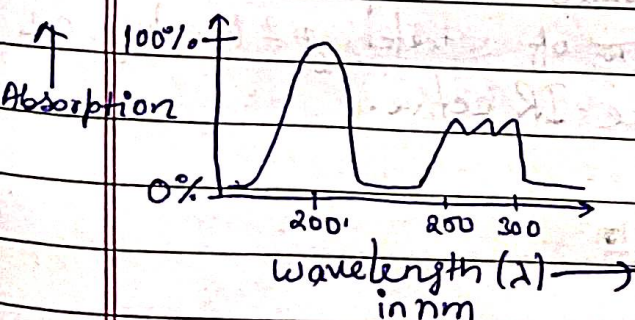


Fig. UV absorption spectrum of benzene in ethanol.

$\pi \rightarrow \pi^*$

180 nm
 200 nm
 255 nm

• Application of UV Spectroscopy Photometer:-

- ① Determination of Conc of unknown sample.
- ② Study of Cis-trans isomers.
- ③ Extent of Conjugation in polymers.
- ④ Determination of vitamin-A in fish-oil.

• IR-SPECTROSCOPY

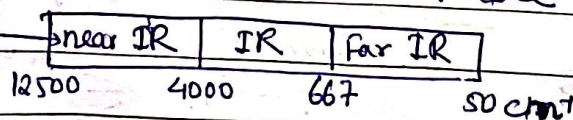
- IR radiation is a part of electromagnetic spectrum between the visible and microwave region.
- It is absorbed by organic molecules and converted into energy of molecular vibration (either stretching or bending).
- Homonuclear diatomic molecules are unable to absorb radiation because they possess zero dipole moment and are thus IR inactive.
- There should be a permanent dipole moment or a partial change in dipole moment for a molecule to be IR active.
- Natural frequency of vibration of the molecule should match with that of infrared radiation.

Q. Is CO_2 IR active?

Ans: ~~Q. Is CO_2 IR active?~~ CO_2 has a zero dipole moment due to its linear geometry but on contrary of expectation, CO_2 is IR active due to developing partial dipole-moment because of stretching and bending.

- Heteronuclear diatomic or polyatomic which possesses permanent or partial dipole moment on account of stretching & bending are able to absorb to IR radiation are IR active.

(Fingerprint region)



→ (Waveno.)

→ The IR region below 1500 cm^{-1} is called fingerprint region which is used to determine identity of two compounds. Some molecules containing the same functional group shows similar absorptions above 1500 cm^{-1} but their spectra differ in fingerprint region.

• Fundamental Vibration modes & types

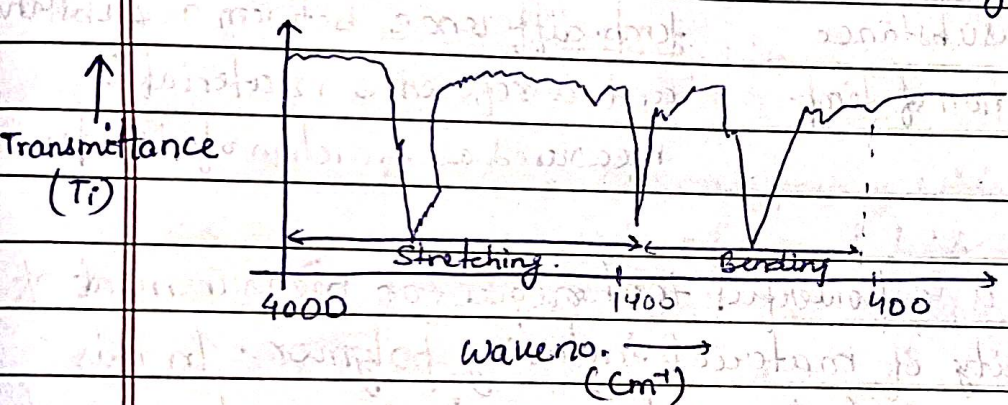
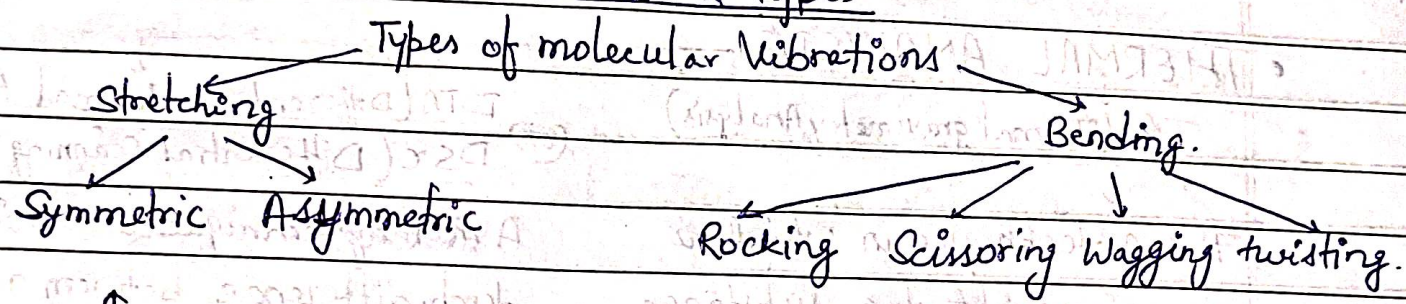
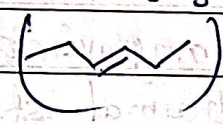


Fig. IR spectra of trans-3-hexene



Q. Write difference between IR & UV ~~spectroscopy~~ Spectroscopy.

- | <u>UV</u> | <u>IR</u> |
|---|--|
| 1) UV visible radiation possesses more energy, they are strong and so can cause electronic excitation. | 1) IR radiations are weak and so cannot produce electronic excitation. They can cause bending & stretching vibration of organic bonds. |
| 2) Unsaturated and especially conjugated groups only absorb and so can cause electronic excitation in this region. | 2) All organic bonds and groups absorb in this region. |
| 3) UV-absorption spectrum exhibits broad absorption bands. | 3) IR spectrum exhibits sharp peaks. |

4) ~~IR~~ IR-Spectra are usually expressed in terms of nm-unit.

4) are usually expressed in terms of wave no. unit (cm^{-1}).

→ A molecule like HCl can undergo stretching vib. only while a molecule like CO_2 can undergo stretching as well as bending vibratⁿ.

• THERMAL ANALYSIS

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TGA (Thermal gravimetry Analysis)

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It is a technique in which a change in weight of a substance is recorded as a function of temp. or time.

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DTA (Differential Thermal Analysis)

(oo) DSC (Differential Scanning Calorimetry)

↓
A ~~tech~~ technique in which the temp. difference between a substance and a reference material is measured as function of temp.

→ Thermogravimetry is a powerful technique for measurement of the thermal stability of materials including polymer. In this method, change in weight of a substance is measured as function of temperature.

• CHROMATOGRAPHY:

→ It's a method by which a mixture is separated by disturbing its components b/w two phases.

→ The stationary phase remain fixed in place while mobile phase carries the components of the mixture through the medium on being used.

It is an important biophysical technique that ~~en~~ enable the separation, identification, purification of components of a sample (mixture) for quantification and qualification.

Four types of Chromatography are—

- ① Gas Chromatography (GC) ② High performance liquid chromatography (HPLC)
③ Thin layer chromatography (TLC) ④ Paper chromatography (PC)

Each has its own advantages and benefits in several industries from healthcare to forensic science.

Q. Write short notes on ^{GC}GLC and HPLC and their differences.

① Gas-liquid Chromatography :- ~~Quantitative method~~

It is a column chromatographic technique useful to separate components in mixtures that easily vapourise without decomposition. In this method, the mobile phase is a carrier gas while stationary phase is a liquid. Usually carrier gas is an inert gas like He or N₂.

② HPLC :- Components of sample interacts with liquid phase.

It is a column chromatography which included pumping the sample in solvent into column at high pressure. Here stationary phase is a solid absorbent. The components of sample interacts with solid stationary phase.

• Surface Characterisation techniques :-

→ It means the properties associated with its surface. It typically measures Surface Area, Surface roughness, pore size. Constitutes Surface characterisation.

Classical Method ← SCT → Industrial Method

(SEM) → short notes
TEM

• Scanning Electron Microscope :-

It transmit a beam which focuses at the specific and scan each line of the surface of the sample in rectangular pattern and provides 3D image of high resolution. These image provides informatⁿ for topology and morphology. It is used in research and quality control.